Putting it all together - Work-Energy Theorem.

For each example below, fill in the table (overpage) with the required quantities

A A man lifts a 5 kg rucksack from the floor to a hook, 2m above the ground. The rucksack starts and ends at zero speed. ***System = only the rucksack***

B Same problem but ***system = rucksack + earth***

C A girl drags a 10 kg sandbag along the horizontal floor for a distance of 2.5 m applying a constant horizontal pulling force of 8N on the sandbag. Friction between sandbag and ground is assumed to be negligible. Sandbag initially stationary. ***System = the sandbag***.

D The girl drags the 10 kg sandbag along the horizontal floor for a distance of 2.5 m applying a constant horizontal pulling force of 8N on the sandbag, maintaining the sandbag at a constant speed of 1.0 m/s. ***System = the sandbag.***

E The girl drags the 10 kg sandbag along the horizontal floor for a distance of 2.5 m applying a constant horizontal pulling force of 8N on the sandbag. Average friction is assumed to be 2N. Sandbag initially stationary. ***System = the sandbag.***

F The girl drags the 10 kg sandbag up an inclined plane, increasing its height by 10 cm, for a distance of 2.5 m applying a constant horizontal pulling force of 8N. Average friction is assumed to be 2N. Sandbag initially moving at 1.0 m/s. ***System = the sandbag + Earth.***

G The girl drags a 10 kg sandbag up an inclined plane, increasing its height by 10 cm, for a distance of 2.5 m applying a constant horizontal pulling force of 8N. Average friction is assumed to be 2N. Sandbag initially moving at 1.0 m/s. ***System = the sandbag.***

H A 65 kg box is pushed up a frictionless slope inclined at 30° to the horizontal, applying a force of 600 N (parallel to the surface), over a distance of 2 m. ***System = the box***

I Same problem, but ***system = the box + the Earth***

J A 65 kg box is pushed up a slope inclined at 30° to the horizontal, applying a force of 600 N (parallel to the surface), over a distance of 2 m. The average frictional force is 5 N . ***System = the box***

K Same problem, but ***system = the box + the Earth***

L A spring with spring constant 2.5N/cm is preloaded by compressing to half its normal length (which is 8.0 cm) It is used to fire a ball, mass 0.1 kg. ***System = ball***

M Same problem, but ***system = ball + spring***

N A spring mounted lever on a pin-ball machine is used to accelerate a ball as it is ascending a slope. The ball, mass 45 g, is moving up the slope at a speed of 0.64 m/s when the spring mounted lever releases, accelerating the ball up the slope. A certain time later the ball has increased in height by 8.5 cm and is moving at a speed of 1.2 m/s. During this time the ball has moved a distance of 55 cm and has experienced an average frictional force of 0.14 N. ***Take the system to be the earth, the ball and the spring***

O Same problem, but take the ***system to be only the ball.***

The table shows the quantities as described in the work-energy theorem

(if it helps, for Ug, put (Ug)i = 0, then (Ug)f = ΔUg

	All in joules						
Situ	K_i	U_{gi}	U_{si}	W	K_f	U_{gf}	U_{sf}
A							
B							
C							
D							
E							
F							
G							
H							
I							
J							
K							
L							
M							
N							
O							

Notes:

The origin of energy to provide positive external work can be from a variety of sources (e.g. chemical, electrical, etc.)

When we have negative work contributions, this implies that energy is leaving the system and therefore increasing somewhere else. For example: the height of an object in a gravitational field or the internal energy (production of heat).